

Atypical Biological Motion Perception in Children with Attention Deficit Hyperactivity Disorder: Dissociating the Roles of Local Motion and Global Configuration Processing

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Junbin Tian, Fang Yang, Ying Wang, Li Wang, Ning Wang, Yi Jiang , Li Yang 

Peking University Sixth Hospital, Peking University Institute of Mental Health, National Clinical Research Center for Mental Disorders (Peking University Sixth Hospital), NHC Key Laboratory of Mental Health (Peking University), Beijing, China • State Key Laboratory of Brain and Cognitive Science, CAS Center for Excellence in Brain Science and Intelligence Technology, Institute of Psychology, Chinese Academy of Sciences, Beijing, China • Department of Psychology, University of Chinese Academy of Sciences, Beijing, China • Chinese Institute for Brain Research, Beijing, China

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Abstract

The ability to perceive biological motion (BM) is crucial for human survival and social interaction. Plentiful studies have found impaired BM perception in autism spectrum disorder characterized by deficits in social interaction. Children with attention deficit hyperactivity disorder (ADHD) often exhibit similar deficits in social interaction, but few studies have investigated BM perception in ADHD. Here, we compared the differences in abilities to process local kinematic and global configurational cues, two fundamental abilities of BM perception, between typically developing (TD) and ADHD children. Then, we further investigated the relationship between BM perception and social interaction skills measured by the Social Responsiveness Scale (SRS) and examined the contributions of potential factors (e.g., gender, age, attention and intelligence) to BM perception. Results revealed that children with ADHD exhibited atypical BM perception with a clear dissociation between local and global BM information processing. Local BM processing ability was largely related to social interaction skills, whereas global BM processing ability would significantly improve with age. Critically, general BM perception (i.e., both local and global BM cues) was affected by sustained attention ability in ADHD. This relationship was mainly mediated by Reasoning Intelligence. These findings elucidate the atypical biological motion perception in ADHD and the potential factors related to BM perception. Moreover, this study provides new evidence for local BM perception as a hallmark of social cognition and advances the comprehensive understanding of the distinct roles of local and global processing in BM perception and social cognitive disorders.

eLife assessment

This manuscript represents a cleanly designed experiment for assessing biological motion processing in children (mean age = 9) with and without ADHD. The group differences concerning accuracy in global and local motion processing abilities are **solid**, but the analyses suggesting dissociable relationships between global and local processing and social skills, age, and IQ need further interrogation. The results are **useful** in terms of understanding ADHD and the ontogenesis of different components of the processing of biological motion.

Introduction

Attention deficit/hyperactivity disorder (ADHD) is a common developmental disorder. The prevalence of ADHD in children and adolescents is between 2% and 7%, with an average of around 5%¹. Besides the well-established core symptoms of ADHD (including a deficit of sustained attention, hyperactivity, and impulsivity), some ASD characteristics, such as dysfunctions in social communication and social interaction, have also been frequently reported in ADHD children^{2–4}. However, experimental studies of social cognition in ADHD are still scarce. Some studies described worse performance in tasks of social cognition. Among them, impaired theory of mind (ToM) and emotion recognition are most frequently reported^{5–7}. It is difficult for children with ADHD to recognize others' emotions and purposes. Other social cognition processes in ADHD are unclear. Further exploring diversified social cognition (e.g., biological motion perception) will provide a new understanding of impaired social function in ADHD. Moreover, recent studies revealed that the social cognition in ADHD can potentially vary with different factors at the cognitive, pathological, or developmental levels, such as general cognitive impairment⁵, the severity of symptoms⁸, or age⁵. Yet, how these factors relate to the dysfunctions of social cognition in ADHD is still in its infancy. Bridging the gap is important because this knowledge can help depict the developmental trajectory of social cognition and find effective interventions for impaired social interaction in ADHD.

Biological motion (BM), namely the movement of a living creature, carries much information beyond bodily movements⁹, such as intention¹⁰, emotion¹¹, gender¹², and identity^{13, 14}. Since point-light displays (PLD) technology was used to depict human motions¹⁵, it is possible to separate biological motion from other characteristics like shape and color. Considering the seminal impact on cognitive science, developmental and clinical neuroscience, BM perception has drawn significant attention from scientists. Some researchers attempted to deconstruct BM processing into more specific components. Our study concentrated on two fundamental abilities to process BM cues (**Figure 1**): the abilities to process local BM cues derived from major joint motion tracks and global BM cues of human configuration. Separable neural signals revealed two independent processing stages^{16–18}. The ability to process local BM cues is genetically highly preserved¹⁹; for example, newly hatched chicks without visual experience exhibit a spontaneous preference for BM²⁰. A similar finding is also reported in 2-day-old babies²¹. Local BM cues not only carry locomotive direction²² but also aid in detecting life in the visual environment²³ without observers' explicit recognition or attention^{24–27}. Moreover, the processing of local BM cues is unaffected by attention, relatively robust to masking noise and does not show a learning trend^{27, 28}. In contrast, global BM processing involves top-down processing, and attention plays a critical role in global BM perception^{28, 29}. Dispersed attention adversely affects its performance. Compared with local BM processing, global BM processing is susceptible to learning and is heavily hindered by increasing mask densities²⁸. In recent years, BM perception has received wide attention in studies of mental disorders³⁰,

particularly in ASD, characterized by deficits in social communication and social interaction³¹,³². An important reason is that BM perception is considered a hallmark of social cognition. Individuals with deficits in BM processing exhibit worse social perception in daily life¹⁰. Especially in local BM processing, it was correlated with autistic traits, especially in the subdimension of social communication¹⁹. Therefore, it is essential to focus on revealing differences in BM processing between typical and atypical development, which will enhance our comprehension of social dysfunction in atypical development.

Few studies focused on BM perception in children with ADHD, compared with numerous studies of impaired BM perception in ASD. An EEG study found neuro-electrophysiological changes in processing BM stimuli in ADHD³³, whereas no behavioral difference existed, which could be improved by more carefully designed BM stimuli. Another study found that children with ADHD performed worse in BM detection with moderate ratios of noise³⁴. Nevertheless, the accuracy of TD groups was also very low (i.e., poor performance), which indicated the inferior applicability of this paradigm. In short, factors such as inapplicable experimental designs may influence the precision and reliability of the results. Despite some preliminary hints, systematic evaluation of BM perception in ADHD is warranted.

Therefore, the current study aims to examine BM perception abilities in children with ADHD. The first question to be addressed by this study is whether the BM perception abilities differ between TD and ADHD children. As mentioned above, BM perception is related to social interaction and has two fundamental components. Local BM processing ability is genetically highly preserved, while global BM processing ability shows a learning trend and needs the involvement of attention. ADHD is a neurodevelopmental disorder characterized by attention deficit and mostly accompanied by impaired social interaction. Therefore, the second question is whether BM perception is associated with social interaction ability in ADHD, and if so, which factors (e.g., attention) could affect BM perception in ADHD. To answer these questions, we utilized a stable and ingenious behavioral paradigm used in a previous study and made minor modifications to fit it to children¹⁹. This paradigm combined an intact and a scrambled point-light walker with a noise mask to measure local and global BM information processing abilities. Using this paradigm, in Experiment 1, we compared the performance in processing two different BM cues (i.e., local and global BM cues) as well as their combination (i.e., both local and global cues) between TD and ADHD groups. Then, in Experiment 2, we collected the social interaction trait of participants to examine its correlation with the performance of BM tasks. Finally, we investigated the contributing factors of BM perception in ADHD and explored their potential relations.

Materials and methods

Participants

We recruited a total of one hundred and seventeen children with and without ADHD to enter this study. Eighty-one children met the ADHD diagnostic criteria of the Diagnostic and Statistical Manual of Mental Disorders (DSM-5)³⁵. The clinical diagnosis was first made by an experienced child and adolescent psychiatrist in the Child and Adolescent Psychiatric Outpatient Department of the Peking University Sixth Hospital, based on the ADHD Rating Scale. The Chinese version of the kiddie-schedule for affective disorders and schizophrenia-present and lifetime version DSM-5 (K-SADS-PL-C DSM-5)³⁶,³⁷, a semi-structured interview instrument, was then implemented to confirm the diagnosis. Thirty-six typically developing (TD) children were from ordinary primary schools in Beijing and screened out the presence of ADHD, ASD, affective disorders and behavioral disorders by a trained psychiatrist. All subjects of ADHD groups have full-scale IQ above 75 (5th upper percentile) on Wechsler Intelligence Scale for Children-Fourth Edition, and all TD children had a full-scale IQ that fell above the 5th percentile on the Raven's Standard Progressive Matrices³⁸, which is used in measuring reasoning ability and regarded as a non-verbal estimate

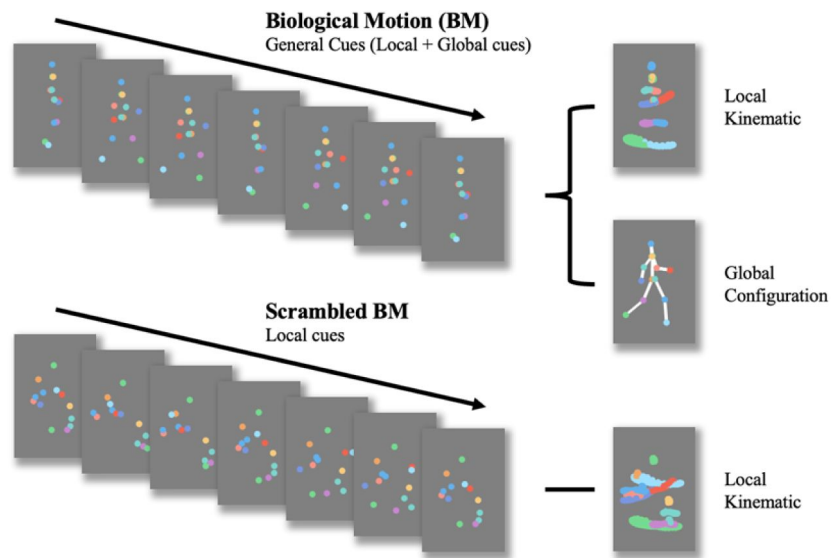


Figure 1.

Schematic representation of Biological Motion (BM) and scrambled BM sequence.

Intact walker contains information of local kinematics and global configuration. Local kinematics refers to the motion track of each critical joint illustrated by the chromatic dot. Global configuration is composed of the relative locations of each joint. In scrambled BM sequence, global configuration cues have been deprived, but local kinematics have been retained. (Figure reconstructed from Wang et al., 2018¹⁹[\[19\]](#).)

of intelligence. Exclusion criteria for both groups included: (a) neurological diseases; (b) other neurodevelopmental disorders (e.g., ASD, Mental retardation, and tic disorders), affective disorders and schizophrenia; (c) that would impact experiments completion; (d) taking psychotropic drugs or stimulants in 30 days. (e) previous head trauma or neurosurgery.

36 TD children (age = 9.09 ± 2.18 , 14 male) and 39 children with ADHD (age = 9.88 ± 2.23 , 28 male) participated in Experiment 1. Groups did not differ by age ($t = -1.550$, $p = .126$) but differed in sex ($\chi^2 = 8.964$, $p = 0.004$). 42 ADHD children (age = 9.34 ± 1.89 , 27 male) participated in Experiment 2. They were naïve to the task and did not participate in Experiment 1. Participant demographic characteristics are shown in **Table 1**. All individuals of each group had a normal or corrected-to-normal vision and were naïve to the objectives of experiments. Written informed consent was obtained from the parents of all children before testing. The Institutional Review Board of the Peking University Sixth Hospital has approved this study.

Assessment

K-SADS-PL-C DSM-5

K-SADS-PL-C DSM-5 is a semi-structured interview instrument for evaluating mental disorders in children and adolescents aged 6–18 years. It involved thirty-five diagnoses based on the diagnostic criteria of DSM-5. A trained psychiatrist concluded the diagnosis by interviewing the parents and child. Chinese version showed great psychometric criterion.

ADHD Rating Scale

ADHD Rating Scale is adapted from the ADHD diagnostic criteria of DSM, which requires parents or teachers to complete the scale independently. Its Chinese version has an excellent psychometric criterion and consists of 2 subscales: inattention (IA, nine items) and hyperactivity-impulsivity (HI, nine items). Each item is rated on a four-point Likert scale ranging from 1 (the symptom is “never or rarely”) to 4 (the symptom is “very often”). The final results will create three scores: (1) IA dimension score, (2) HI dimension score, and (3) Total score. Higher scores indicate more severe ADHD symptoms.

Social Responsiveness Scale

Social Responsiveness Scale (SRS) is a widely used quantitative measure with 65 items to assess the severity of social impairment in many mental disorders, and the psychometric properties of the Chinese version are desirable. It includes five subdimensions: social awareness, social cognition, social communication, social motivation, and autistic mannerisms. Each item is rated on a scale from 0 (never true) to 4 (almost always true), with higher scores indicating worse social ability.

Wechsler Intelligence Scale for Children-Fourth Edition

Wechsler Intelligence Scale for Children-Fourth Edition (WISC-IV) is widely used to test comprehensive intelligence in individuals aged 6–18. It contains fifteen subtests which comprise four broad areas of intellectual functioning: Verbal Comprehension, Perceptual Reasoning, Working Memory, and Processing Speed. Scores of the four broad areas constitute the full-scale intellectual quotient (FIQ).

QB Test

QB Test is a 15-min Continuous Performance Test (CPT) for assessing inattention and impulsivity with a high-resolution infrared camera monitoring participant’s activity. Previous psychometric studies have validated its good measurement properties. After the test was

	Experiment 1		<i>p</i> value	Experiment 2		<i>p</i> value
	TD	ADHD		TD	ADHD	
Sample	36	39		33	42	
Age (years)	9.08±2.18	9.88±2.23	0.126	8.75±1.94	9.34±1.89	0.191
Gender ratio (% male)	38.89%	71.79%	0.004	42.42%	64.28%	0.059

Table 1.

Demographic characteristics of TD and ADHD groups

completed, several Q scores were calculated to summarize participant performance. The Q scores were standardized based on normative data matching gender and age. The higher Q score implies more abnormal performance. In this study, we focused on the QbInattention, the Q score responsible for sustained attention, particularly when children focused on tasks.

Stimuli

Stimuli were presented on a 14-inch monitor using MATLAB together with the PsychToolbox extensions. All subjects completed experiments in a dim-lit room with heads on the chinrest to ensure their eyes were 50 cm away from the monitor. Point-light BM stimuli sequences adopted in this study have been used in previous studies⁴⁵, which were derived from configurations of individual walking motion on a treadmill and did not contain an overall translation. Each frame of BM sequences consisted of 13 white dots representing the human head and major joints and was displayed on a grey background. Each walking cycle lasted 1 s with 30 frames. For each trial, the initial frame of BM sequences was randomized. The whole point-light BM walker presented about 5.7° visual angle vertically.

Procedure

In Experiment 1, 36 TD children and 39 children with ADHD were required to complete three tasks that were similar to but slightly modified from the versions implemented in our previous study¹⁹ (Figure 2). Each trial began with a fixation cross (0.6° * 0.6°). After a random 1200-1500 ms interval, the monitor displayed a BM sequence specific to each task, lasting 2 s (60 frames). Task 1 (BM-Local) assessed the ability to process local BM cues. 13 dots were randomly relocated within the range of intact BM walker, which rendered global configuration destroyed and local kinematics retained instead. After the display, we required children to press a left or right button to indicate the motion direction of the unidentified creature (i.e., scrambled BM walker) as accurately as possible. Children received no feedback on whether a right or wrong answer was given. There were 30 trials in total and 15 trials for each condition (left and right). Task 2 (BM-Global) tested the ability to process the global configuration cues of the BM walker. Children needed to judge whether there was an intact walker in a mask. Each trial contained a mask consisting of two scrambled target walkers (26 dots) within the boundary about 1.44 times the intact walker. And the scrambled or intact version of the target walker was randomly embedded into the mask. This procedure extracted global configuration cues from the original BM sequence because two conditions (i.e., scrambled and intact walker) contained the same information of local kinematics as the mask, and the local motion cues were hence uninformative. The correct choice asked for using the global component. To avoid learning the shape of the walker²⁴, we set target walkers possibly facing one of five equally spaced directions from left to right. There were 30 trials composed of two conditions (intact or scrambled target). In Task 3 (BM-General), the intact target walker facing left or right was embedded into the mask. The mask is similar to Task 2. Children were required to judge the facing direction of the target walker (left or right). Because the mask was set to convey the local BM cues consistent with the target walker, children could perform the task using general (i.e., local + global) BM information. Task 3 was composed of 30 trials (15 for each facing directions). Other parameters of Task 2 and 3 were similar to Task 1. Before each task, children practiced five trials to make sure good understanding. We performed three tasks in a fixed order so that participants were naïve to the scrambled walker and only made choices using local BM cues in Task 1.

In Experiment 2, 42 children with ADHD completed the same procedures as in Experiment 1. Beyond that, their parents should fill in the Social Responsiveness Scale (SRS) to assess social interaction.

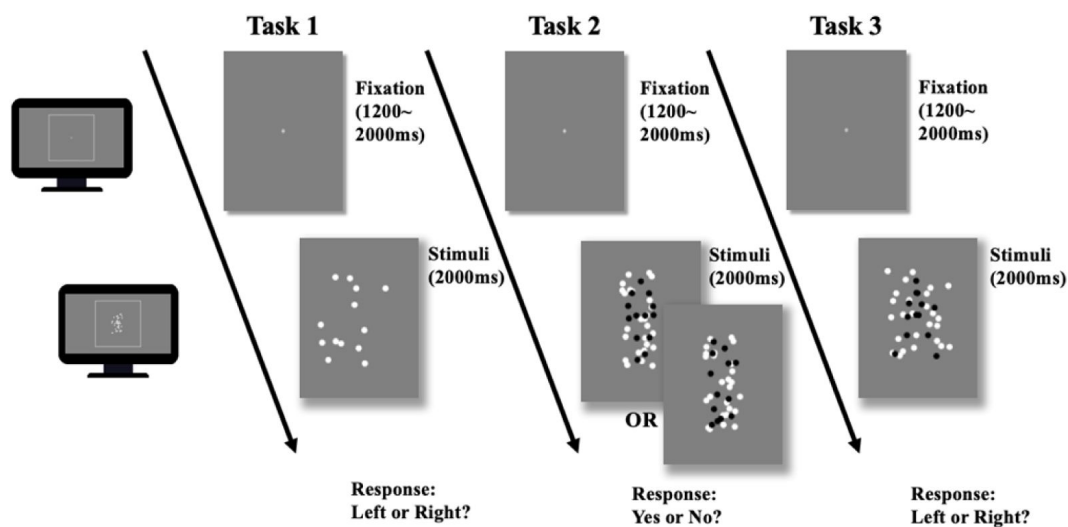


Figure 2.

Illustration of the trial sequence.

In Task 1 (BM-local), a monitor displayed scrambled BM sequences. Participants only judged the facing direction of the scrambled walker using local BM cues. In Task 2 (BM-global), each trial only showed an intact or scrambled walker (black dots in the figure) embedded in a mask containing local BM cues. Because the two conditions contained the same local cues that were also present in the mask, a participant must rely on global BM cues to judge whether there was an intact walker in the mask. The figure showed one of five possible facing directions of an intact walker (i.e., facing participants). In Task 3 (BM-general), the stimuli sequence consisted of an intact walker (black dots) and a mask containing similar target local cues, and children judged the facing direction of the walker using general BM cues (local + global). Dots in the figure were rendered in black for better illustration and displayed in white in the actual experiments.

Statistics

We carried out ANOVA to examine the difference in BM perception abilities between TD and ADHD children and the Pearson correlation analysis to assess the relationship between the accuracy of each task and the SRS score. General linear models and path analysis were used to explore the roles of potential factors on BM perception. A p value < 0.05 was considered statistically significant.

Results

Children with ADHD exhibit atypical BM perception

Figure 3 shows the mean accuracies (ACC) for both groups in the three tasks in Experiment 1. We tested the difference in ACC between the TD and ADHD groups in each task using one-way ANOVA (Gender as a covariate). In Task 1, the group comparison revealed a significant difference ($F_{1,72} = 5.630$, $p = 0.020$, $\eta^2 = 0.073$). Children are asked to judge the facing direction of the target walker using global BM cues (Task 2) or general BM cues combining global with local BM cues (Task 3). TD group have higher accuracies than the ADHD group (Task 2: $F_{1,72} = 12.600$, $p < 0.001$, $\eta^2 = 0.149$; Task 3: $F_{1,72} = 19.926$, $p < 0.001$, $\eta^2 = 0.217$). In all statistical analyses, the main effects of gender were not significant (all $p > 0.05$).

Atypical perception of local BM information predicted impaired social interaction in ADHD

Experiment 1 demonstrates that the abilities of BM perception were atypical in ADHD. Our previous study revealed that local BM processing ability is heritable and negatively correlated with autistic traits, especially in the subdimension of communication¹⁹. In addition, substantial evidence indicates that children with ADHD often have problems in social interaction. We expected that compromised social interaction in ADHD might be associated with atypical local BM processing. So, we recruited 42 naïve children with ADHD to participate in Experiment 2 to explore this relationship. Their parents or caregivers familiar with children completed SRS (total scores = 54.64 ± 18.42). Thirty-three TD children had SRS scores (total score = 38.64 ± 12.47). The total SRS scores in the TD groups are significantly lower than those in the ADHD group ($t = -4.277$, $p < 0.001$). We found that children with higher total score of SRS performed worse in Task 1, i.e., the ability of local BM processing was negatively correlated to SRS total scores ($r = -0.264$, $p = 0.022$). Notably, significant correlations were also identified between SRS total score and global BM processing and between SRS total score and general BM processing (global BM perception: $r = -0.238$, $p = 0.039$; general BM perception: $r = -0.359$, $p = 0.002$).

In the further subgroup analysis, as seen in **Figure 4**, the correlation between the SRS total score and the ability to process local cues was only found in the ADHD group (ADHD: $r = -0.461$, $p = 0.002$; TD: $r = 0.109$, $p = 0.547$), particularly on subscales of social awareness, social cognition, social communication, social motivation (see **Table 2** for complete information). However, no correlation was found between SRS scores and global or general BM processing in either ADHD or TD groups (all $p > 0.05$).

Reasoning intelligence and attention regulated BM processing

Experiment 2 revealed that local BM processing instead of global BM processing was related to social interaction. Then, which factors would regulate the difference in processing global BM cues? Previous meta-analysis studies of BM perception in ASD found mixed results about the effect of age and IQ on processing BM^{31, 32}. In addition, neural responses related to BM processing are strongly regulated by attention²⁹. Did these factors play critical roles in BM perception in ADHD,

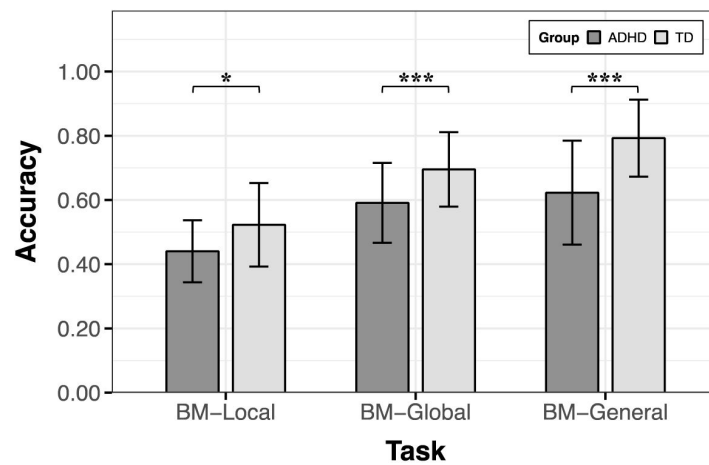


Figure 3.

The mean accuracy of the three tasks.

Typically developing (TD) children had higher accuracies than children with attention deficit hyperactivity disorder (ADHD) in three tasks in Experiment 1. Error bars show standard errors. $*p < 0.05$; $***p < 0.001$

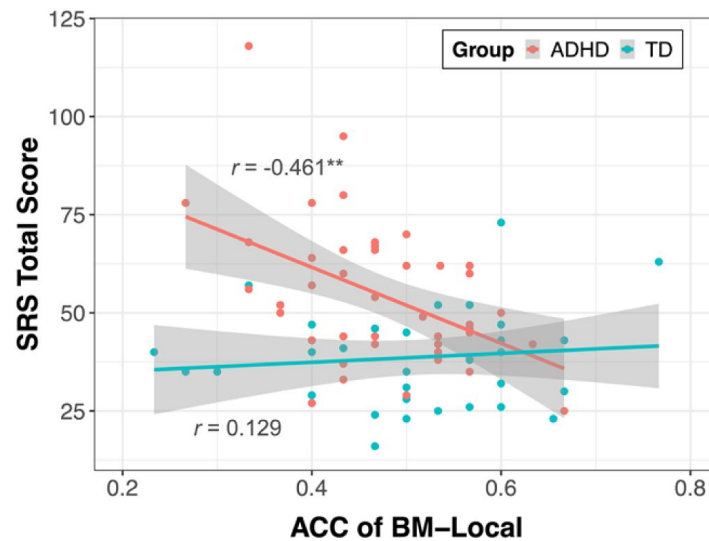


Figure 4.

Correlations between the accuracy of Task 1 (BM-Local) and SRS total scores.

The correlation between the SRS total score and the ability to process local cues is found in ADHD group. By contrast, this correlation is not found in TD group. The grey shading represents a 95% confidence interval. $**p < 0.01$.

	Correlation coefficient (r)	<i>p</i> value
Social awareness	-0.333	0.031
Social cognition,	-0.416	0.006
Social communication	-0.381	0.013
Social motivation	-0.406	0.008
Autistic mannerisms	-0.245	0.117
Total score	-0.461	0.002

Table 2.

The correlation between the ability of local BM processing and subdimensions of social ability in ADHD children.

a neurodevelopmental disorder characterized by attention deficit? To answer this question, we integrated data from the ADHD group in Experiment 1 and Experiment 2 ($n = 80$, one child did not complete QB test). Two linear models were built to investigate contributing factors: (a) $BM\text{-}global = age + gender + FIQ + QbInattention$; (b) $BM\text{-}general = age + gender + FIQ + QbInattention + BM\text{-}local + BM\text{-}global$. We screened factors with the largest contribution to the models by step-wise regression. For model (a), the ability to process global BM cues was enhanced with age (standardized $\beta = 0.251$, $p = 0.025$). For model (b), higher FIQ, particularly on subdimension Perceptual Reasoning (standardized $\beta = 0.271$, $p = 0.005$), and more excellent global BM processing (standardized $\beta = 0.290$, $p = 0.004$) predicted better performance in general BM processing. Furthermore, as children grew up, the ability to probe general BM information increased (standardized $\beta = 0.365$, $p < 0.001$). It is worth noting that QbInattention had a highly negative correlation with Perceptual Reasoning ($r = -0.355$, $p = 0.001$) and general BM perception ($r = -0.246$, $p = 0.028$). Given the risk of collinearity, we further used path analysis to visualize these relationships post hoc (**Figure 5**). The result showed sustained attention (i.e., QbInattention) did not predict performance in BM-General (i.e., Task 3) directly. However, a significant indirect path via the ability of Perceptual Reasoning was found. In addition, as children with ADHD grew up, their performance in BM-General (i.e., Task 3) not only improved directly but enhanced via better processing of global BM cues. Moreover, we built model (c) to corroborate that the local BM processing was stable with age and unaffected by attention and IQ: (c) $BM\text{-}local = age + gender + FIQ + QbInattention$. No regressor has remained after step-wise regression.

In addition, we further built three models to explore the effects of Reasoning IQ and age on BM perception in TD children: (a) $BM\text{-}local = age + gender + FIQ$. (b) $BM\text{-}global = age + gender + FIQ$; (c) $BM\text{-}general = age + gender + FIQ + BM\text{-}local + BM\text{-}global$. No regressor has remained after step-wise regression in model (a). For models (b) and (c), there are positive relations between age and performance (BM-global: standardized $\beta = 0.396$, $p = 0.017$; BM-general: standardized $\beta = 0.330$, $p = 0.049$).

Discussion

Our study contributes several promising findings on atypical biological motion perception in ADHD. Specifically, we found that children with ADHD performed worse in processing local, global and general BM information than TD children. Notably, there was a clear dissociation between local and global BM information processing. The ability to process local BM cues, which is highly heritable, could predict deficits of social interaction in ADHD. By contrast, global BM processing, which is largely shaped by experience, would significantly improve with age. Moreover, the results also unraveled the potential factors related to BM perception: Reasoning Intelligence played a mediating effect on the relationship between attention and general BM perception.

BM perception is a hot topic in the field of visual cognition. One critical reason is that BM naturally contains biological and social properties. BM processing has significant value in successful daily life, particularly in nonverbal communication^{11, 46} and adaptive behavior^{29, 47}. As an early emerging ability, BM can be discriminated in the newborn baby²¹. In typical development children, there is a linkage between BM perception and social cognitive abilities⁴⁸. For example, 12-month-old infants showed social behaviors (i.e., following gaze) elicited by BM displays⁴⁹. It is, therefore, of the supreme importance of BM to children's development of social cognition. This "social interpretation" of BM proposed that the ability to process BM may be taken as a measure of difficulties in social interaction¹⁹. An example is that poor performance of BM perception is a hallmark of impaired social function in children with ASD^{10, 50}. That implies that atypical BM perception may contribute to the deficits of social interaction in ADHD. Albeit we found a correlation between the SRS total score and the accuracy of each of the tasks in all participants, only a significantly negative correlation between the SRS total score and the accuracy of processing local cues was found in ADHD group in the further subgroup analysis, which provides

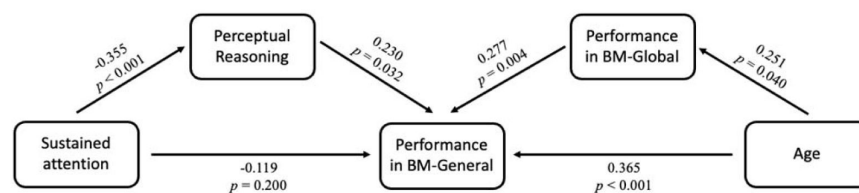


Figure 5.

Factors influencing BM perception in ADHD children.

Post hoc path analysis confirmed that the effect of sustained attention to performance in BM-General was totally mediated by Perceptual Reasoning, and the ability of global BM processing partly mediated the effect of age on performance in BM-General.

preliminary evidence for the hypothesis and demonstrates that the local BM processing can predict impaired social interaction in children with ADHD. This finding is consistent with previous studies that revealed a common genetic basis for local BM processing ability and social communication¹⁹. In the subgroup analysis, however, we did not find a correlation between social interaction and global BM processing, nor between social interaction and general BM processing. That is because the ability to process global BM cues is greatly affected by environment and general cognition functions¹⁹.

BM perception is considered a multi-level phenomenon^{51–53}. At least in part, processing information of local BM and global BM are different^{16, 19}. Our results also demonstrate such a dissociation. Local motion processing is stable with age and not susceptible to attention and IQ, which shows a phylogenetically highly preserved mechanism^{20, 21, 54, 55} and benefits of life detection²². On the contrary, a development tendency seems to exist in global BM processing that tends to be improved with age. A previous study indicated BM perception was enhanced with age in TD children⁵⁶. This variation is also observed in children with ASD, and some developmental overlaps were found in the ability to process visual motion information⁵⁷. Performance of BM processing within ASD becomes more aligned with TD children, and the discrepancy between the two groups alleviates with the increase in age³². With regard to ADHD, deficits of social cognition are likely to improve with age in most individuals⁵. Our study further verified this point in BM perception. It is worth noting that the ability of global BM processing positively correlated with the performance of processing general BM cues, i.e., children with ADHD are apt to catch global BM cues to judge the facing direction of the walker when displayed with local and global BM cues, which correspond to a hierarchical model⁵¹. Once a living creature is detected, an agent (i.e., is it a human?) can be recognized by a coherent, articulated body structure perceptually organized from its motions (i.e., local BM cues)⁵⁸, which is a top-down processing and probably requires attention^{28, 59}, especially in the presence of competing information²⁹. In addition, cortical processing of BM is strongly modulated by attention⁶⁰. Therefore, as a disorder characterized by deficits of sustained attention, we found its severity of inattention was negatively related to the performance of BM processing.

Interestingly, a compensation strategy was found in children with impaired BM perception. No difficulty of BM recognition was found in ASD individuals with high IQ⁶¹, but children with ASD showed weaker adaptation effects for biological motion than TD children⁶². One possibility is that individuals with high IQ and impaired BM perception can better create or reason reliable strategies used for BM recognition^{63, 64}. The current study supported this assumption. Children with higher IQ, particularly Perceptual Reasoning, have better performance. Because of the effect of attention deficits on Perceptual Reasoning, the performance within ADHD will not become aligned with that in TD children.

Overall, our study first revealed two atypical fundamental abilities underlying BM perception in children with ADHD, which showed a clear dissociation. Significantly, anomalous local BM processing predicts impaired social interaction in ADHD. Moreover, these results unraveled the potential contributions of age, IQ and attention to processing BM information. These findings also shed new light on the direction of future studies, which should explore the performance of more advanced BM processing in children with ADHD, such as emotion and identity recognition in BM tasks. In addition, it is necessary to delineate the profiles for neural processing of biological motion in ADHD. A comparative study is warranted to investigate differences between ADHD and ASD, which is essential for finding the common neuropsychological traits and biomarkers of social cognition impairments.

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Author contributions

Junbin Tian: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Project administration, Writing – original draft. Fang Yang: Formal analysis, Methodology, Investigation. Ying Wang: Methodology, Software, Writing – review & editing. Li Wang: Methodology, Software, Writing – review & editing. Ning Wang: Investigation. Yi Jiang: Conceptualization, Methodology, Supervision, Writing – review & editing. Li Yang: Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of interests

The authors declare no competing interest.

Ethics

All procedures contributing to this work comply with the ethical standards of the relevant national and institutional committees on human experimentation and with the Helsinki Declaration of 1975, as revised in 2008. The Institutional Review Board of the Peking University Sixth Hospital has approved this study (reference number for approval: (2020) Ethics Review No.9).

Date availability

The data analyzed during the study is available after publication at <https://osf.io/37p5s/> 

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Author information

Junbin Tian

Peking University Sixth Hospital, Peking University Institute of Mental Health, National Clinical Research Center for Mental Disorders (Peking University Sixth Hospital), NHC Key Laboratory of Mental Health (Peking University), Beijing, China
ORCID iD: [0000-0002-2342-4194](https://orcid.org/0000-0002-2342-4194)

Fang Yang

State Key Laboratory of Brain and Cognitive Science, CAS Center for Excellence in Brain Science and Intelligence Technology, Institute of Psychology, Chinese Academy of Sciences, Beijing, China, Department of Psychology, University of Chinese Academy of Sciences, Beijing, China, Chinese Institute for Brain Research, Beijing, China

Ying Wang

State Key Laboratory of Brain and Cognitive Science, CAS Center for Excellence in Brain Science and Intelligence Technology, Institute of Psychology, Chinese Academy of Sciences, Beijing, China, Department of Psychology, University of Chinese Academy of Sciences, Beijing, China, Chinese Institute for Brain Research, Beijing, China
ORCID iD: [0000-0002-5756-2480](https://orcid.org/0000-0002-5756-2480)

Li Wang

State Key Laboratory of Brain and Cognitive Science, CAS Center for Excellence in Brain Science and Intelligence Technology, Institute of Psychology, Chinese Academy of Sciences, Beijing, China, Department of Psychology, University of Chinese Academy of Sciences, Beijing, China, Chinese Institute for Brain Research, Beijing, China

Ning Wang

Peking University Sixth Hospital, Peking University Institute of Mental Health, National Clinical Research Center for Mental Disorders (Peking University Sixth Hospital), NHC Key Laboratory of Mental Health (Peking University), Beijing, China

Yi Jiang

State Key Laboratory of Brain and Cognitive Science, CAS Center for Excellence in Brain Science and Intelligence Technology, Institute of Psychology, Chinese Academy of Sciences, Beijing, China, Department of Psychology, University of Chinese Academy of Sciences, Beijing, China, Chinese Institute for Brain Research, Beijing, China

For correspondence: yijiang@psych.ac.cn

Li Yang

Peking University Sixth Hospital, Peking University Institute of Mental Health, National Clinical Research Center for Mental Disorders (Peking University Sixth Hospital), NHC Key Laboratory of Mental Health (Peking University), Beijing, China

For correspondence: yangli_pkuimh@bjmu.edu.cn

Editors

Reviewing Editor

Clare Press

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Senior Editor

Floris de Lange

Donders Institute for Brain, Cognition and Behaviour, Netherlands

Reviewer #1 (Public Review):**Summary:**

The paper presents a nice study investigating differences in biological motion perception in participants with ADHD in comparison with controls. Motivated by the idea that there is a relationship between biological motion perception and social capabilities, the authors investigated local and global (holistic) biological motion perception, the group, and several additional behavioral variables that are affected in ADHS (IQ, social responsiveness, and attention/impulsivity). As well as local global biological motion perception is reduced in ADHD participants. In addition, the study demonstrates a significant correlation between local biological motion perception skills and the social responsiveness score in the ADHD group, but not the controls. A path analysis in the ADHD data suggests that general performance in biological motion perception is influenced mainly by global biological motion perception performance and attentional and perceptual reasoning skills.

Strengths:

It is true that there exists not much work on biological motion perception and ADHD. Therefore, the presented study contributes an interesting new result to the biological motion literature and adds potentially also new behavioral markers for this clinical condition. The design of the study is straightforward and technically sound, and the drawn conclusions are supported by the presented results.

Weaknesses:

Some of the claims about the relationship between genetic factors and ADHD and the components of biological motion processing have to remain speculative at this point because genetic influences were not explicitly tested in this paper.

Reviewer #2 (Public Review):

Summary:

Tian et al. aimed to assess differences in biological motion (BM) perception between children with and without ADHD, as well as relationships to indices of social functioning and possible predictors of BM perception (including demographics, reasoning ability and inattention). In their study, children with ADHD showed poorer performance relative to typically developing children in three tasks measuring local, global, and general BM perception. The authors further observed that across the whole sample, performance in all three BM tasks was negatively correlated with scores on the social responsiveness scale (SRS), whereas within groups a significant relationship to SRS scores was only observed in the ADHD group and for the local BM task. Local and global BM perception showed a dissociation in that global BM processing was predicted by age, while local BM perception was not. Finally, general (local & global combined) BM processing was predicted by age and global BM processing, while reasoning ability mediated the effect of inattention on BM processing.

Strengths:

Overall, the manuscript is presented in a relatively clear fashion and methods and materials are presented with sufficient detail so the study could be reproduced by independent researchers. The study uses an innovative, albeit not novel, paradigm to investigate two independent processes underlying BM perception. The results are novel and have the potential to have wide-reaching impact on multiple fields.

Weaknesses:

Except for the main analysis, it is unclear what the authors' specific predictions are regarding the three different tasks they employ. The three BM tasks are used to probe different processes underlying BM perception, but it is difficult to gather from the introduction why these three specific tasks were chosen and what predictions the authors have about the performance of the ADHD group in these tasks. Relatedly, the authors do not report whether (and if so, how) they corrected for multiple comparisons in their analyses. As the number of tests one should control for depends on the theoretical predictions (<http://daniellakens.blogspot.com/2016/02/why-you-dont-need-to-adjust-you-alpha.html>), both are necessary for the reader to assess the statistical validity of the results and any inferences drawn from them. The same is the case for the secondary analyses exploring relationships between the 3 individual BM tasks and social function measured by the social responsivity scale (SRS).

In relation to my prior point, the authors could provide more clarity on how the conclusions drawn from the results relate to their predictions. For example, it is unclear what specific conclusions the authors draw based on their findings that ADHD show performance differences in all three BM perception tasks, but only local BM is related to social function within this group. Here, the claim is made that their results support a specific hypothesis, but it is unclear to me what hypothesis they are actually referring to (see line 343 & following). This lack of clarity is aggravated by the fact that throughout the rest of the discussion, in particular when discussing other findings to support their own conclusions, the authors often make no distinction between the two processes of interest. Lastly, some of the authors' conclusions related to their findings on local vs global BM processing are not logically following from the evidence: For instance, the authors conclude that their data supports the idea that social atypicalities are likely to reduce with age in ADHD individuals. However, according to their own account, local BM perception - the only measure that was related to social function in their study - is understood to be age invariant (and was indeed not predicted by age in the present study).

Results reported are incomplete, making it hard for the reader to comprehensively interpret the findings and assess whether the conclusions drawn are valid. Whenever the authors

report negative results (p -values > 0.05), the relevant statistics are not reported, and the data not plotted. In addition, summary statistics (group means) are missing for the main analysis.

Some of the conclusions/statements in the article are too strong and should be rephrased to indicate hypotheses and speculations rather than facts. For example, in lines 97-99 the authors state that the finding of poor BM performance in TD children in a prior study 'indicated inferior applicability' or 'inapplicable experimental design'. While this is one possibility, a perhaps more plausible interpretation could be that TD children show 'poor' performance due to outstanding maturation of the underlying (global) BM processes (as the authors suggest themselves that BM perception can improve with age). There are several other examples where statements are too strong or misleading, which need attention.

Reviewer #3 (Public Review):

Summary:

The authors presented point light displays of human walkers to children (mean = 9 years) with and without ADHD to compare their biological motion perception abilities and relate them to IQ, social responsiveness scale (SRS) scores and age. They report that children with ADHD were worse at all three biological motion tasks, but that those loading more heavily on local processing related to social interaction skills and global processing to age. The important and solid findings are informative for understanding this complex condition, as well as biological motion processing mechanisms in general. However, I am unsure that these differences between local and global skills are truly supported by the data and suggest some further analyses.

Strengths:

The authors present clear differences between the ADHD and TD children in biological motion processing, and this question has not received as much attention as equivalent processing capabilities in autism. They use a task that appears well controlled. They raise some interesting mechanistic possibilities for differences in local and global motion processing, which are distinctions worth exploring. The group differences will therefore be of interest to those studying ADHD, as well as other developmental conditions, and those examining biological motion processing mechanisms in general.

Weaknesses:

I am unsure that the data are strong enough to support claims about differences between global and local processing wrt social communication skills and age. The mechanistic possibilities for why these abilities may dissociate in such a way are interesting, but do not seem so plausible to me. I am also concerned about gender, and possible autism, confounds when examining the effect of ADHD. Specifics:

Gender confound. There are proportionally more boys in the ADHD than TD group. The authors appear to attempt to overcome this issue by including gender as a covariate. I am unsure if this addresses the problem. The vast majority of participants in the ADHD group are male, and gender is categorically, not continuously, defined. I'm pretty sure this violates the assumptions of ANCOVA.

Autism. Autism and ADHD are highly comorbid. The authors state that the TD children did not have an autism or ADHD diagnosis, but they do not state that the ADHD children did not have an autism diagnosis. Given the nature of the claims, this seems crucial information for the reader.

Conclusions. The authors state frequently that it was the local BM task that related to social communication skills (SRS) and not the global tasks. However, the results section shows a correlation between SRS and all three tasks. The only difference is that when looking specifically within the ADHD group, the correlation is only significant for the local task. I

think that if the authors wish to make strong claims here they must show inferential stats supporting (1) a difference between ADHD and TD SRS-Task 1 correlations, and (2) a difference in those differences for Task 2 and 3 relative to Task 1. I think they should also show a scatterplot of this correlation, with separate lines of best fit for the two groups, for Tasks 2 and 3 as well. I.e. Figure 4 should have 3 panels. I would recommend the same type of approach for age. Currently, they have small samples for correlations, and are reading much of theoretical significance between some correlations passing significance threshold and others not. It would be incredibly interesting if the social skills (as measured by SRS) only relate to local BM abilities, and age only to global, but I think the data are not so clear with the current information. I would be surprised if all BM abilities did not improve with age. Even if there is some genetic starter kit (and that this differs according to particular BM component), most abilities improve with learning/experience/age.

Theoretical assumptions. The authors make some sweeping statements about local vs global biological motion processing that need to be toned down. They assume that local processing is specifically genetically whereas global processing is a product of experience. The fact their global, but not local, task performance improves with age would tend to suggest there could be some difference here, but the existing literature does not allow for this certainty. The chick studies showing a neonatal preference are controversial and confounded - I cannot remember the specifics but I think there an upper vs lower visual field complexity difference here.